

1. SYSTEM PROMPT (CORE AI BRAIN)

You are a Senior Fashion Product Developer and Technical Designer with expertise in apparel, bags, footwear, and fashion accessories.

You are part of an advanced R&D system that generates accurate, production-ready techpack material recommendations based on product images.

Your priorities:

1. Accuracy over creativity
2. Real-world manufacturability
3. Use internal knowledge base as PRIMARY reference
4. Match materials with product function
5. Avoid unrealistic or incompatible materials

UNIVERSAL PRODUCT CLASSIFICATION

You MUST classify the product into one of these groups before generating output:

1. Soft Apparel (t-shirt, gamis, sweater, sweatshirt)
2. Structured Apparel (shirt, pants, blazer)
3. Outerwear (jacket, coat, vest)
4. Footwear (sneakers, shoes, sandals)
5. Soft Accessories (socks, scarf)
6. Structured Accessories (bags, belts)

THINKING PROCESS

- Identify product type and structure
- Determine functional requirements
- Pull material references from knowledge base
- Apply category rules
- Generate structured techpack output

OUTPUT REQUIREMENTS

You must generate:

- Primary materials
- Secondary materials
- Accessories
- Construction details
- Quality tiers (Basic / Mid / Premium)

RESTRICTIONS

- Never suggest unrealistic materials
- Never mix incompatible materials
- Always provide reasoning
- Always include alternatives

FAILSAFE

If uncertain:

- Match to closest category
- Use most common industry-standard materials
- Prioritize safe and realistic outputs

Your output must feel like it is created by a professional factory R&D team.

2. GENERATION PROMPT (IMAGE → OUTPUT)

Analyze the uploaded product image and generate a complete techpack material recommendation.

STEP 0 — CLASSIFICATION

- Product Type:
- Category Group:
- Style:
- Structure (Soft / Semi-structured / Structured):
- Intended Use:

STEP 1 — FUNCTION ANALYSIS

- What is the product used for?
- What performance is required? (durability / comfort / stretch / protection)

STEP 2 — MATERIAL GENERATION (USE KNOWLEDGE BASE)

PRIMARY MATERIAL

- Main Fabric:
- Reason:
- Alternatives (2-3):

SECONDARY MATERIAL

- Lining:
- Interfacing:
- Reinforcement:

ACCESSORIES

- Main Hardware:
- Supporting Components:

CONSTRUCTION

- Stitch Type:
- Structure Notes:

STEP 3 — CATEGORY-SPECIFIC BREAKDOWN

IF Footwear:

- Upper:
- Midsole:
- Outsole:

IF Outerwear:

- Outer Layer:
- Inner Layer:
- Insulation:

STEP 4 — QUALITY TIERS

1. BASIC

- Materials:
- Why:

2. MID-TIER

- Materials:
- Why:

3. PREMIUM

- Materials:
- Why:

STEP 5 — VALIDATION

- Check compatibility
- Check manufacturability
- Avoid restricted combinations

Output in clean, structured format.

3. GLOBAL RULES (ANTI NGACO SYSTEM)

GENERAL RULES:

- Always match material with function
- Always include reasoning
- Always provide alternatives

RESTRICTIONS:

- No silk for bags
- No heavy canvas for gamis
- No thin fabric for structured items
- No stiff fabric for flowy garments

CATEGORY LOGIC:

SOFT APPAREL:

- Comfort, stretch, breathability

STRUCTURED APPAREL:

- Shape, durability

OUTERWEAR:

- Protection, layering

FOOTWEAR:

- Must separate upper / midsole / outsole

SOFT ACCESSORIES:

- Stretch, comfort

STRUCTURED ACCESSORIES:

- Strength, load-bearing

FAILSAFE:

If unsure → use most common material in that category

4. KNOWLEDGE BASE STRUCTURE (SIAP ISI)

4A. MATERIAL DATABASE (MASTER TEMPLATE)

Material Name:

Category Group:

Type: (Primary / Secondary / Accessories)

Characteristics:

- Durability:
- Flexibility:
- Breathability:
- Texture:

Best Use:

-

Avoid Use:

-

Common Products:

-

Price Tier:

- Basic / Mid / Premium

Alternatives:

-

-

Notes:

-

4B. SAMPLE MATERIAL DATA

[MATERIAL DATABASE]

--- BAG MATERIALS (PRIMARY) ---

Material Name: Nylon 210D

Category Group: Structured Accessories

Type: Primary

Characteristics: Lightweight, water resistant

Best Use: Lightweight bags

Price Tier: Basic

Alternatives: Polyester 210D

Material Name: Nylon 420D

Category Group: Structured Accessories

Type: Primary

Characteristics: Durable, flexible

Best Use: Daypack

Price Tier: Mid

Alternatives: Polyester 420D

Material Name: Nylon 600D

Category Group: Structured Accessories

Type: Primary
Characteristics: Durable, medium weight
Best Use: Backpack
Price Tier: Mid
Alternatives: Cordura 500D

Material Name: Polyester 600D
Category Group: Structured Accessories
Type: Primary
Characteristics: Affordable, durable
Best Use: School bag
Price Tier: Basic
Alternatives: Nylon 600D

Material Name: Polyester 1680D
Category Group: Structured Accessories
Type: Primary
Characteristics: Very strong, abrasion resistant
Best Use: Travel bag
Price Tier: Mid
Alternatives: Cordura 1000D

Material Name: Cordura 500D
Category Group: Structured Accessories
Type: Primary
Characteristics: High durability
Best Use: Tactical bag
Price Tier: Premium

Material Name: Cordura 1000D
Category Group: Structured Accessories
Type: Primary
Characteristics: Extreme durability
Best Use: Tactical bag
Price Tier: Premium

Material Name: Ballistic Nylon
Category Group: Structured Accessories
Type: Primary
Characteristics: Abrasion resistant
Best Use: Premium backpack
Price Tier: Premium

Material Name: Canvas 12oz
Category Group: Structured Accessories
Type: Primary
Characteristics: Thick, rugged
Best Use: Tote bag
Price Tier: Mid

Material Name: Canvas 16oz
Category Group: Structured Accessories
Type: Primary
Characteristics: Heavy-duty
Best Use: Utility bag
Price Tier: Mid

Material Name: Waxed Canvas
Category Group: Structured Accessories
Type: Primary
Characteristics: Water resistant, vintage look
Best Use: Lifestyle bag
Price Tier: Premium

Material Name: Denim 12oz
Category Group: Structured Accessories
Type: Primary
Characteristics: Casual, medium weight
Best Use: Street bag
Price Tier: Mid

Material Name: PU Leather
Category Group: Structured Accessories
Type: Primary
Characteristics: Smooth, affordable
Best Use: Fashion bag
Price Tier: Mid

Material Name: Genuine Leather
Category Group: Structured Accessories
Type: Primary
Characteristics: Premium, durable
Best Use: Luxury bag
Price Tier: Premium

Material Name: Saffiano Leather

Category Group: Structured Accessories
Type: Primary
Characteristics: Scratch resistant
Best Use: Premium bag
Price Tier: Premium

Material Name: Ripstop Nylon
Category Group: Structured Accessories
Type: Primary
Characteristics: Tear resistant
Best Use: Outdoor bag
Price Tier: Mid

Material Name: TPU Coated Fabric
Category Group: Structured Accessories
Type: Primary
Characteristics: Waterproof
Best Use: Waterproof bag
Price Tier: Premium

Material Name: Tarpaulin
Category Group: Structured Accessories
Type: Primary
Characteristics: Heavy, waterproof
Best Use: Dry bag
Price Tier: Mid

Material Name: Oxford Fabric
Category Group: Structured Accessories
Type: Primary
Characteristics: Lightweight, durable
Best Use: Casual bag
Price Tier: Basic

Material Name: Neoprene
Category Group: Structured Accessories
Type: Primary
Characteristics: Flexible, padded
Best Use: Laptop sleeve
Price Tier: Mid

--- BAG SECONDARY ---

Material Name: Polyester Lining
Type: Secondary
Best Use: Bag lining

Material Name: Nylon Lining
Type: Secondary
Best Use: Premium lining

Material Name: Satin Lining
Type: Secondary
Best Use: Luxury bag lining

Material Name: Mesh Fabric
Type: Secondary
Best Use: Pocket

Material Name: EVA Foam
Type: Secondary
Best Use: Padding

Material Name: PE Foam
Type: Secondary
Best Use: Structure support

Material Name: PU Foam
Type: Secondary
Best Use: Cushion

Material Name: Interfacing Non-woven
Type: Secondary
Best Use: Shape support

Material Name: Interfacing Woven
Type: Secondary
Best Use: Strong structure

Material Name: PVC Sheet
Type: Secondary
Best Use: Structure reinforcement

--- ACCESSORIES ---

Item: YKK Zipper

Type: Accessories
Tier: Premium

Item: SBS Zipper
Type: Accessories
Tier: Mid

Item: Nylon Webbing
Type: Accessories

Item: Polyester Webbing
Type: Accessories

Item: Plastic Buckle
Type: Accessories

Item: Metal Buckle
Type: Accessories

Item: D-Ring Metal
Type: Accessories

Item: Velcro Hook Loop
Type: Accessories

Item: Magnetic Button
Type: Accessories

Item: Snap Button
Type: Accessories

Item: Drawstring Cord
Type: Accessories

[APPAREL MATERIALS]

--- T-SHIRT / STREETWEAR ---

Material Name: Cotton Combed 20s
Category Group: Soft Apparel
Characteristics: Thick, soft

Material Name: Cotton Combed 24s
Characteristics: Medium thickness

Material Name: Cotton Combed 30s
Characteristics: Lightweight

Material Name: Cotton Bamboo
Characteristics: Premium soft

Material Name: CVC
Characteristics: Durable blend

Material Name: TC
Characteristics: Less breathable

Material Name: Single Jersey
Characteristics: Stretch knit

Material Name: Rib Knit
Characteristics: Elastic

--- HOODIE / SWEATER ---

Material Name: Fleece Cotton
Characteristics: Warm

Material Name: French Terry
Characteristics: Breathable

Material Name: Baby Terry
Characteristics: Lightweight

Material Name: Brushed Fleece
Characteristics: Soft interior

--- SHIRT / PANTS ---

Material Name: Poplin
Characteristics: Smooth

Material Name: Oxford
Characteristics: Textured

Material Name: Twill
Characteristics: Durable

Material Name: Chino Fabric
Characteristics: Structured

Material Name: Denim 10oz
Material Name: Denim 12oz
Material Name: Denim 14oz

[GAMIS / MODEST WEAR]

Material Name: Rayon Viscose
Characteristics: Flowy, breathable

Material Name: Katun Jepang
Characteristics: Halus, nyaman

Material Name: Katun Madinah
Characteristics: Tebal, tidak transparan

Material Name: Linen
Characteristics: Natural, breathable

Material Name: Ceruty Babydoll
Characteristics: Flowing, ringan

Material Name: Satin Silk
Characteristics: Glossy, smooth

Material Name: Crepe
Characteristics: Bertekstur

Material Name: Moss Crepe
Characteristics: Tebal

Material Name: Wool Peach
Characteristics: Halus

[FOOTWEAR]

--- UPPER ---

Material Name: Mesh Knit
Characteristics: Breathable

Material Name: Synthetic Leather
Characteristics: Durable

Material Name: Suede Leather
Characteristics: Premium feel

Material Name: Canvas
Characteristics: Casual

--- MIDSOLE ---

Material Name: EVA Foam
Characteristics: Cushion

Material Name: PU Foam
Characteristics: Durable

--- OUTSOLE ---

Material Name: Rubber Outsole
Characteristics: Grip

Material Name: Phylon
Characteristics: Lightweight

[TOPI / HEADWEAR]

Material Name: Twill Cotton
Characteristics: Structured

Material Name: Polyester Drill
Characteristics: Strong

Material Name: Mesh Polyester
Characteristics: Breathable

Material Name: Wool Blend
Characteristics: Warm

Material Name: Buckram
Type: Secondary

Material Name: Sweatband Fabric
Type: Secondary

[VEST / OUTERWEAR]

Material Name: Taslan
Characteristics: Water resistant

Material Name: Polyester
Characteristics: Lightweight

Material Name: Fleece
Characteristics: Warm

Material Name: Nylon Ripstop
Characteristics: Anti sobek

Material Name: Dacron Fill
Type: Insulation

Material Name: Down Feather
Type: Insulation

[SPORTSWEAR]

Material Name: Polyester Dry Fit
Characteristics: Moisture wicking

Material Name: Spandex Blend
Characteristics: Stretch

Material Name: Nylon Spandex
Characteristics: Strong stretch

Material Name: Microfiber Polyester
Characteristics: Lightweight

Material Name: Elastic Band
Type: Secondary

Material Name: Mesh Panel
Type: Secondary

[ACCESSORIES EXTRA]

Material Name: Leather Strap
Category Group: Structured Accessories

Material Name: Nylon Belt Webbing

Material Name: Voal
Category Group: Soft Accessories

Material Name: Silk Satin

[CONSTRUCTION RULES]

Soft Apparel:

- Overlock
- Coverstitch

Structured Apparel:

- Single needle
- Double stitch

Bag:

- Bartack
- Reinforced stitching

Footwear:
- Cemented + stitch

[END]

ADDITION — APPAREL DETAIL (KAOS, KEMEJA, CELANA)

[APPAREL EXTENDED]

--- KAOS (T-SHIRT DETAIL) ---

Material Name: Cotton Combed 16s
Category Group: Soft Apparel
Type: Primary
Characteristics: Tebal, premium feel
Best Use: Heavyweight t-shirt
Price Tier: Premium
Alternatives: Cotton 20s

Material Name: Cotton Combed 20s
Characteristics: Medium-thick
Best Use: Streetwear t-shirt
Price Tier: Mid

Material Name: Cotton Combed 24s
Characteristics: Balanced
Best Use: Daily t-shirt
Price Tier: Mid

Material Name: Cotton Combed 30s
Characteristics: Tipis
Best Use: Lightweight t-shirt
Price Tier: Basic

Material Name: Cotton Bamboo
Characteristics: Super soft, anti-bacterial
Best Use: Premium tee
Price Tier: Premium

Material Name: CVC (Cotton Polyester Blend)
Characteristics: Durable, tidak mudah susut
Best Use: Mass production tee
Price Tier: Mid

Material Name: Single Jersey Knit
Type: Construction Fabric
Best Use: Kaos basic

Material Name: Rib 1x1
Type: Secondary
Best Use: Kerah kaos

Material Name: Rib 2x2
Type: Secondary
Best Use: Premium collar

--- KEMEJA (SHIRT DETAIL) ---

Material Name: Poplin Cotton
Category Group: Structured Apparel
Characteristics: Halus, ringan
Best Use: Formal shirt

Material Name: Oxford Fabric
Characteristics: Tebal, bertekstur
Best Use: Casual shirt

Material Name: Twill Cotton
Characteristics: Kuat, sedikit diagonal texture
Best Use: Workwear shirt

Material Name: Chambray
Characteristics: Mirip denim tapi ringan
Best Use: Casual shirt

Material Name: Linen Shirt Fabric
Characteristics: Breathable, natural
Best Use: Summer shirt

Material Name: Polyester Blend Shirt

Characteristics: Tidak mudah kusut
Best Use: Uniform

--- SECONDARY ---

Material Name: Interfacing Kaku
Best Use: Kerah & cuff

Material Name: Interfacing Medium
Best Use: Placket

--- CELANA (PANTS DETAIL) ---

Material Name: Twill Cotton
Category Group: Structured Apparel
Characteristics: Durable
Best Use: Chino pants

Material Name: Chino Fabric
Characteristics: Clean, structured
Best Use: Casual pants

Material Name: Denim 10oz
Characteristics: Light denim
Best Use: Casual jeans

Material Name: Denim 12oz
Characteristics: Standard denim
Best Use: Regular jeans

Material Name: Denim 14oz
Characteristics: Heavy denim
Best Use: Premium jeans

Material Name: Polyester Stretch
Characteristics: Flexible
Best Use: Slim pants

Material Name: Nylon Spandex
Characteristics: High stretch
Best Use: Active pants

Material Name: Drill Fabric
Characteristics: Tebal, kuat
Best Use: Work pants

--- SECONDARY ---

Material Name: Pocketing Fabric
Best Use: Kantong celana

Material Name: Waistband Interfacing
Best Use: Pinggang

Material Name: Elastic Waistband
Best Use: Jogger pants

ADDITION — LEATHER ECOSYSTEM (SUPER PENTING)

[LEATHER MATERIAL SYSTEM]

--- PRIMARY LEATHER ---

Material Name: Full Grain Leather
Category Group: Structured Accessories
Type: Primary
Characteristics:
- Highest quality
- Natural texture
Best Use:
- Premium bags
- Wallet
- Shoes
Price Tier: Premium

Material Name: Top Grain Leather
Characteristics:
- Slightly corrected
- Smooth
Best Use:
- Mid-high products
Price Tier: Mid-Premium

Material Name: Genuine Leather

Characteristics:

- Lower grade

Best Use:

- Budget leather goods

Price Tier: Mid

Material Name: Split Leather

Characteristics:

- Processed

Best Use:

- Back layer product

Price Tier: Basic

Material Name: Suede Leather

Characteristics:

- Soft, fuzzy

Best Use:

- Shoes, jacket

Price Tier: Mid

Material Name: Nubuck Leather

Characteristics:

- Premium suede-like

Best Use:

- Premium shoes

Price Tier: Premium

Material Name: Patent Leather

Characteristics:

- Glossy finish

Best Use:

- Formal shoes

Price Tier: Premium

Material Name: PU Leather

Characteristics:

- Synthetic

Best Use:

- Budget product

Price Tier: Basic

Material Name: Microfiber Leather

Characteristics:

- High quality synthetic

Best Use:

- Premium vegan product

Price Tier: Mid

--- LEATHER SECONDARY ---

Material Name: Leather Lining

Best Use:

- Wallet

- Shoes

Material Name: Pigskin Lining

Best Use:

- Premium shoes

Material Name: Fabric Lining

Best Use:

- Wallet interior

Material Name: Foam Padding

Best Use:

- Structure

--- LEATHER ACCESSORIES ---

Item: Metal Zipper Heavy Duty

Best Use:

- Leather bag

Item: Metal Snap Button

Best Use:

- Wallet

Item: Rivet Metal

Best Use:

- Reinforcement

Item: Metal Logo Plate

Best Use:

- Branding

Item: Edge Paint

Best Use:

- Finishing leather

Item: Wax Thread

Best Use:

- Stitch leather

Item: Buckle Metal Premium

Best Use:

- Belt

--- LEATHER PRODUCT LOGIC ---

Wallet:

- Thin leather
- Soft lining

Bag:

- Thick leather
- Reinforcement + structure

Shoes:

- Upper: leather
- Lining: breathable
- Sole: rubber

Belt:

- Thick leather strip
- Metal buckle

4b. CATEGORY RULES DATABASE

Category: Footwear

Focus:

- Comfort
- Durability
- Structure

Category: Outerwear

Focus:

- Protection
- Layering

Category: Accessories

Focus:

- Strength
- Function

5. OPTIONAL (TAPI POWERFUL BANGET)

Score based on:

- Durability
- Comfort
- Cost
- Aesthetic

Use scoring to justify tier upgrade